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 HSS504
 May 14, 2026

Cochran Wildlife Sanctuary Value Assessment

Throughout this assessment, we use a discount rate of 2.5%. Currently, the Trump administration does not have a discount rate, so we used the rates from the Biden administration. The Biden administration used a 2% discount rate, which increased the social cost of carbon to \$190 in 2023. Biden's 2023 SCC was far higher than previous administrations' due to its use of a much lower discount rate compared to previous rates of between 3-7%.¹ Our group ultimately decided on a 2.5% discount rate, in order to account for the medium time horizon of 50 years. This discount rate amounts to a SCC of \$158 as of 2026.² All of the following calculations assume that Phillips Academy Andover intends to maintain ownership of the Sanctuary for at least the next 50 years.

1. Immediate Value

A. Land Value (and Potential Revenue) for Housing Development

The Cochran Wildlife Sanctuary occupies a large plot of land, 65 acres, on our campus in the town of Andover.³ Currently, this land is used for recreational purposes, open to both Andover students and the public. For the purposes of this assignment, our group attempted to evaluate the Sanctuary's immediate land value and performed calculations to determine how much redevelopers would pay based on the potential profit they stand to gain.

Our initial approach began researching how much buildable plots of land retailed for in Andover. However, this search was cut short, due to the minimal amounts of data. We only found two plots, a 2.56 and .57 acre lot, retailing for \$950,000 and \$699,900, respectively.⁴ Given this data limitation, we developed a different strategy for evaluating a more accurate land value for the Sanctuary; we performed calculations to estimate the potential profit for developers, and then calculated how much they would pay for the land itself using the Residual Land Value Method

¹ Environmental and Energy Law Program, Harvard Law School. "The Social Cost of Greenhouse Gases (Carbon Dioxide, Methane, Nitrous Oxide)." Last modified July 29, 2025.
<https://eelp.law.harvard.edu/tracker/the-social-cost-of-carbon/>.

² Institute for Policy Integrity. "Calculating the Social Cost of Greenhouse Gases." *The Cost of Climate Pollution*. Accessed May 13, 2026. <https://costofcarbon.org/calculator>.

³ "Cochran Wildlife Sanctuary," Andover Trails, accessed May 13, 2026,
<https://www.andovertrails.org/cochran-wildlife-Sanctuary.html>.

⁴ "Andover, MA land & lots for sale," Realtors.com, accessed May 13, 2026,
https://www.realtor.com/realestateandhomes-search/Andover_MA/type-land.

(calculated based on the potential profits). All data and constants referred to throughout the section can be seen in the table below.

Calculations:	
Sanctuary in Square Feet:	$65 * 43560 = 2831400$ sq feet
Amount of land plots in the Sanctuary:	$2631400 / 30,056 = 94.2$ homes
Adjustment for Undevelopable Land:	$94.2 * (17/19) = 84.3$ homes
Profit from selling homes (multiply by median list price):	$84.3 * 959900 = \$80,316,600$
Cost of clearcutting:	$11200 * (65 * (17/19)) = \651368.42
Cost of construction:	$323077 * 84.3 = \$27,235,391.1$
Cost to build and clearcut:	$\$651368.42 + \$27,235,391.1 = \$27,886,759.52$
Net Profit:	$\$80,316,600 - \$27,886,759.52 = 52,429,840.48$
Constants:	
Sanctuary Acreage:	65 Acres
Andover House Median list price:	\$959,900
Andover House Median sale price:	\$1,000,000
Andover House Median lot sq footage:	30,056 sq feet
Conversion of square feet to acres:	43560 sq feet = 1 acres
Developable Sanctuary Acreage:	$65 * (17/19) = 58.1578947368$ acres
Cost of Clearcutting 2 acres of heavily forested land	\$6,600-\$11,200 (range)
National Average Cost of Building a House:	\$323,077

In order to estimate the number of houses a developer could fit, we divided the 65 acres into different plots based on the median Andover house square footage. To start, we converted the 65 acres to square feet and divided the product by the median lot square footage (30,056 sq feet), resulting in space for 94 homes. To account for the undevelopable land, such as the ponds or log cabin that already exist within the Sanctuary, we estimated (with Google Maps scaling tools) that these features, and the hypothetical roads that would have to be constructed if the land were redeveloped, would account for 2/19th of the total Sanctuary acreage. Thus, to find a more accurate estimate, given the undevelopable land, we multiplied the 94 house estimate by $\frac{17}{19}$ (the fraction of developable land), which resulted in around 84 homes. To find the amount of potential revenue from these houses (disregarding cost of clearcutting and construction), we used the less expensive median list price of \$959,900 (taking a more conservative approach) which resulted in \$80,316,600.

This number does not reflect the developer's revenue because it doesn't account for the cost of clear-cutting or construction required to redevelop the land. To calculate these costs, we looked up average quotes for the clearcutting of heavily forested land.⁵ The costs were considerable, with 2 acre lots costing anywhere from \$6,600-11,200 (see figure below). We used the upper cost boundary (\$11,200) to calculate this expense, which resulted in a quota of \$651368.42 to clear all the developable land of the Sanctuary (disregarding the land occupied by ponds and space for roads).

⁵ "How Much Does it Cost to Clear Land in 2026?," Lawn Love, accessed May 13, 2026, <https://lawnlove.com/blog/land-clearing-cost/>.

Cost of clearing heavily forested land	
Clearing a heavily forested property will take more work, equipment, and money. Prices start at \$3,300 for an acre and can go over \$10,000 for 2 acres. See below cost estimates for ¼-acre to 2-acre lots.	
Lot size	Average cost to clear heavily forested land
¼ acre	\$415 – \$700
½ acre	\$660 – \$1,120
¾ acre	\$825 – \$1,400
1 acre	\$1,650 – \$2,800
1 ½ acres	\$2,475 – \$4,200
2 acres	\$3,300 – \$5,600
3 acres	\$6,600 – \$11,200

To calculate the cost of construction, we found the national average of building a house from scratch, \$323,077, and multiplied that by the number of house plots, which resulted in a total construction cost of \$27,235,391.1.⁶

Average building costs	
According to Angi , building a house costs \$323,077 on average. However, this can range from as little as \$100,000 and as much as \$950,000.	
Average cost range	\$138,937 to \$531,039
Minimum	\$100,000
Maximum	\$950,000
National Average	\$323,077

Even with these hidden costs, the net potential redevelopment revenue of the Sanctuary for developers is considerable, coming in at \$52,429,840.48.

Finally, using the estimated potential revenue, we calculated how much developers would be willing to pay for the land of the Sanctuary using the Residual Land Value Method, shown below:

1. Residual Land Value Method
The residual method is the most widely used approach for development sites. It works backwards from the completed development value:
1. Calculate the Gross Development Value (GDV) – what the completed development will sell for
2. Subtract all development costs (construction, professional fees, finance costs, etc.)
3. Subtract the developer's profit requirement (typically 15–20% of GDV)
4. What remains is the residual land value

⁶ "How Much Does It Cost To Build A New House 2026," MySitePlan, accessed May 13, 2026, https://www.mysiteplan.com/blogs/news/cost-to-build-a-house?srltid=AfmBOopDvpwQ3yHOWPjm-04ebTKb45Sf_8tsooaRSOrfpIB-ZbwrcFd8.

Our calculations are shown below:

Residual Land Value Calculations:	
Profit from selling homes (multiply by median list price):	$84.3 * 959900 = \$80,316,600$
Cost to build and clearcut:	$\$651368.42 + \$27,235,391.1 = \$27,886,759.52$
Developers Profit Requirement	$.20 * \$80,316,600 = \$16,063,320$
Residual Land Value	$\$80,316,600 - \$27,886,759.52 - \$16,063,320 = \$36,366,520.48$

Based on our calculations, assuming a 20% profit requirement, a developer would pay around \$36.4 million dollars for just the land of the Sanctuary.

B. Timber Value

To calculate the one-time value of selling the Sanctuary's timber, we needed three things: how many acres could actually be logged, how much wood those acres hold, and what that wood is currently worth. The price we used throughout is called the stumpage price: this is what a logging company pays the landowner for trees while they're still standing. It already factors in the logger's costs to cut and haul the wood, so it represents what PA would actually receive.

The Sanctuary today covers 65 acres, as a portion of the original land has since been developed into housing.⁷ Our analysis of August 2025 satellite imagery showed approximately 89% forest coverage, consistent with $\frac{17}{19}$ of the total land area, with the remaining 2/19 made up of the ponds and open clearings. This gives us roughly 58 harvestable forested acres ($65 \times \frac{17}{19} \approx 58.2$ acres).

According to the 2013 Davey Resource Group Natural Resource Assessment of Cochran Sanctuary, the forest is classified as a mature Northern Hardwood community, with most trees dating back to the 1928 planting or earlier. Their i-Tree carbon analysis found the trees store 2,578 tons of carbon, which confirms there's substantial wood volume on the property.⁸

First, it's important to realize that all forests go through a natural cycle of death and regrowth. Older trees eventually die and decompose while younger ones move in to take their place. Because Cochran has been left largely unmanaged for nearly a century, this process has hummed along completely uninterrupted for decades. The USDA Forest Service estimates that natural mortality claims about 1-2% of a forest's basal area every year.⁹ This essentially means a huge portion of the original planted trees have already died, becoming standing "snags" or falling to the forest floor: none of which carry any real commercial timber value. Meanwhile, the younger trees replacing them are still far too small to be sold.

Second, the Davey report's Risk Tree Assessment identified 187 risk trees throughout the Sanctuary, specifically recommending 93 for removal because they were considered moderate or

⁷ "Cochran Wildlife," Andover Trails.

⁸ Davey Resource Group, "Natural Resource Assessment of Cochran Sanctuary," 2013. ; U.S. Bureau of Labor Statistics. "CPI Inflation Calculator." U.S. Department of Labor. Accessed May 13, 2026. https://www.bls.gov/data/inflation_calculator.htm.

⁹ U.S. Department of Agriculture, Forest Service, "By the Numbers," last modified August 1, 2025, <https://www.fs.usda.gov/about-agency/newsroom/by-the-numbers>.

high risk. Quite a few of these were actually documented as dead back when the assessment was done in 2013. The report also flagged an active hemlock woolly adelgid infestation, which had already killed several mature hemlocks and has likely continued to drag down the overall timber quality as time goes on.

Third, there is the issue of how the trees grew. Since many of the trees in the Sanctuary were originally planted by hand rather than growing naturally, they often have poorer trunk forms. This leads to more knots and a weaker taper, both of which lower the grade quality and the stumpage value of the lumber.

For mature, unmanaged hardwood forests in Massachusetts, the USDA Forest Service reports typical volumes of 3,000-7,000 board feet per acre.¹⁰ We used 3500 board feet per acre as our central estimate: on the lower end of that range, to account for the fact that the Sanctuary hasn't been managed for timber and has documented pest damage that reduces usable volume. This gives us approximately 203,000 board feet total, or 203 MBF (MBF = thousand board feet, the standard unit used in timber sales).

We pulled species data from the Davey report's Forest Stand Delineation maps and matched each species to its current Massachusetts stumpage price using the Q4 2025 UMass/DCR Southern New England Stumpage Survey.¹¹

To estimate the timber volume for each species, we used the Forest Stand Delineation maps and species composition graphs included in the Davey report.¹² We took the dominant species from the major forest stands and weighted them based on the area each stand covers. Because the Davey report didn't provide exact board foot counts, these values are approximations based on the canopy composition shown throughout the assessment.

¹⁰ U.S. Department of Agriculture, Forest Service, Northeastern Forest Experiment Station, *The Timber Resource in Massachusetts* (Upper Darby, PA: U.S. Department of Agriculture, Forest Service, 1956), https://www.fs.usda.gov/ne/newtown_square/publications/resource_bulletins/pdfs/scanned/OCR/ne_timber%20resources%20Massachusetts.pdf.

¹¹ "Massachusetts Timber Prices Q4 2025," The Timber Investor, accessed May 14, 2026, <https://thetimberlandinvestor.com/massachusetts-timber-stumpage-prices/>.

¹² Davey Resource Group, "Natural Resource Assessment of Cochran Sanctuary," 2013.

A. Pollution Reduction Value

Forests and their surrounding ecosystems can help reduce pollution in two main areas: water and air. Trees are the best land cover for water quality maintenance by massively increasing rainfall capture, as well as filtering water before slowly releasing it into streams and rivers.¹³ They can also improve air quality through a variety of ways, including reducing air temperature and thus pollution concentration, or directly removing pollutants from the air.¹⁴ However, the stream that runs through Cochran Sanctuary, Rogers Brook, is not used for any drinking purposes. Instead, the town of Andover and Phillips Academy use Haggetts Pond for drinking water.¹⁵ Therefore, we will be primarily focusing on the value of the Sanctuary's air pollution removal services.

To quantify the air pollution removal value of Cochran Sanctuary, we use data from Nowak et al. (2014),¹⁶ which utilizes satellite imagery for forest cover and national meteorological data on hourly wind speed and pollution concentrations to calculate air pollution removal of all forests in the US by state. Given the technical complexity and data availability limitations of replicating their modeling, we use the Nowak study's average annual pollution removal rates for the state of Massachusetts. This introduces several assumptions and adjustments, detailed below.

- a. Forest density adjustment: Nowak's 2010 data calculated a Massachusetts forest cover average of 70%, whereas our own analysis of August 2025 satellite imagery of the Sanctuary provided higher estimates of approximately 89% forest coverage (about $\frac{17}{19}$ of the land area). To account for this, we apply a scaling factor of 89/70 to Nowak's per hectare pollution removal value.
- b. Urban categorization: The Nowak model separated land into "urban" and "rural" categories using 2010 Census data classifications. We chose to use Nowak's "urban" pollution removal value due to the town of Andover being a high-density suburban area. However, this means the true pollution reduction levels in tonnes of pollutants removed is likely higher in reality, as the tree density would be greater than an urban area. On the other hand, urban areas had higher monetary value of the pollution removal, as Nowak used avoided health costs to estimate the monetary value of pollution removal. Since Andover's population density is likely lower than an urban area, the monetary value of pollution removal is likely slightly lower than Nowak's value.
- c. Changes in weather patterns and pollution concentration: Nowak's figures use 2010 meteorological conditions, but since then, global pollutant concentrations and weather patterns (by extension wind speeds) may have changed. This would have an unknown

¹³ Austa Somvichian-Clausen, "The Important Relationship between Forests and Water," American Forests, <https://www.americanforests.org/article/the-important-relationship-between-forests-and-water/>.

¹⁴ Air Pollution Removal by Urban Forests," National Park Service, U.S. Department of the Interior, last modified January 7, 2022, <https://www.nps.gov/articles/000/uerla-trees-air-pollution.htm>.

¹⁵ Andover MA Town Government, Who Is Public Works and How Does the Water Distribution and Sewer Collection Fit In?, <https://andoverma.gov/DocumentCenter/View/11012/Water-Sewer-PDF-5202022>.

¹⁶David J. Nowak et al., "Tree and Forest Effects on Air Quality and Human Health in the United States," *Environmental Pollution* 193 (October 2014): <https://doi.org/10.1016/j.envpol.2014.05.028>.

effect on our value calculation; however, we will assume these conditions have not changed sufficiently for Nowak's figures to be unusable.

- d. Discount rate and time horizon: as with all other value calculations in our estimations, we use a discount rate of 2.5% and a time horizon of 50 years.

With these assumptions and adjustments in mind, below are our calculations:

In 2010, the monetary value of pollution removal by trees in Massachusetts was \$286.30 per hectare per year. In 2026 dollars, this is \$433.56 per hectare per year.

$$\text{Current Value} = \text{Original Value} \times \frac{2026 \text{ CPI}}{2010 \text{ CPI}} = \$286.30 \times \frac{330.213}{218.056} = \$433.56$$

To adjust for the difference in tree coverage: $\$433.56 \times \frac{89}{70} = \551.24 per hectare per year

Multiply by the land area of Cochran Sanctuary for total annual value:
 $\$551.24 \times 65ha = \35830.60 per year

To find the cumulative value over 50 years:

$$\text{Net Present Value} = \sum_{t=1}^{50} \frac{35,830.60}{(1.025)^t} = \$1,016,238.64$$

B. Carbon Removal Value

According to the Davey report, the Sanctuary stores 2,578 tons of carbon and sequesters 85 tons of carbon per year.¹⁷ It is important to note, however, that these numbers are from 2013, so the following calculations and conclusions assume they remain consistent in 2026. Davey estimates the annual carbon reduction at \$55,000, which, when adjusted for inflation, is approximately \$77,792 today.¹⁸ However, these 2013 estimates use a discount rate of approximately 3%, which is higher than our selected discount rate of 2.5%. The current administration does not have a set discount rate or SCC, so we chose our discount rate based on our current knowledge of the environment and the policies surrounding it. Additionally, we wanted to calculate the benefits over 50 years rather than just annually. Therefore, we took the liberty of recalculating the Sanctuary's carbon-reduction benefits over 50 years.

Using our group's discount rate of 2.5%, the EPA estimates that the social cost of carbon would be approximately \$158.¹⁹ The social cost of carbon is an estimate of the environmental harm caused by releasing carbon into the atmosphere. Therefore, when we calculate using the social cost of carbon, we are calculating the damage saved/prevented. Using our discount rate and the social cost of carbon, the SCC for the carbon stored would be \$407,324, and the SCC for the carbon sequestered would be \$13,430. In total, the SCC for the carbon stored and sequestered would be \$420,754. To calculate the carbon-reduction benefits over the next 50

¹⁷ Davey Resource Group, "Natural Resource Assessment of Cochran Sanctuary," 2013.

¹⁸ Davey Resource Group, "Natural Resource Assessment of Cochran Sanctuary," 2013.

¹⁹ Institute for Policy Integrity, "Calculating the Social Cost of Greenhouse Gases."

years, we used the equation $(PV) = FV/[1 + \text{discount rate}]^{\text{(number of years)}}$. Our final equation looked like this: $420,754 = FV/(1 + 0.025)^{(50)}$. After rearranging the equation, we found the future value of the carbon reduction benefits to be \$1,446,177. Put more simply, keeping the Sanctuary would mean avoiding \$1,446,177 in environmental damage.

C. Recreational Use Value

To determine the Recreational Use Value, our group sent out a survey to Andover students filled with various questions gauging how much they value the features of the Sanctuary. **In full transparency, we've taken out some outliers in our survey responses that heavily skewed the data, likely due to misinterpreting survey prompts (ex responding that each individual would pay \$500,000 to use the Sanctuary as an athletic space per year or \$60,000 per year to maintain the Sanctuary as a space for creativity.)*

SURVEY AVERAGES INDIVIDUAL COSTS:	In \$
How much do you think individuals would pay to use the sanctuary as an athletic space as it currently exists (the option to run, walk, etc) (\$ value per year)	115.72
What do you believe the value the of sanctuary is as an educational space (have you gone into the sanctuary for classes and did you find those experiences valuable and enriching?) (\$ value per year)	502.4
How much do you think individuals would pay to use the sanctuary as a space for creativity as it currently exists (art, photography, etc.) (\$ value per year)	113.9591667
How much do you think individuals would pay to use the sanctuary as a place to relax and decompress? (\$ value per year)	104.48
How much do you think individuals would pay to use the sanctuary as a social space as it currently exists? (\$ value per year)	84
How much do you value the privacy of the Andover campus that the sanctuary affords us, how much would you be willing to pay to preserve this (\$ value per year)	153.6
TOTAL SUM	1074.159167

In the survey questions above, we've asked individuals to respond with how much they would be willing to pay per year for each of the Sanctuary's services. To calculate the Recreational Use Value, we've averaged each question's responses and then taken the sum of the averages to determine the total amount a PA student would be willing to pay per year for all the Sanctuary's recreational services. We then multiplied it by the total number of PA students (1200 students, assuming around 300 per grade) to estimate how much PA students value ($1200 * 1074.16 = \$1,288,992$, the total recreational value of the Sanctuary per year). Then, to calculate the cumulative recreational value of the Sanctuary over 50 years, we summed the discounted value per year:

$$\text{Net Present Value} = \sum_{t=1}^{50} \frac{\$1,288,992}{(1.025)^t} = \$36,558,792.86$$

Thus, our total Recreational Use Value over 50 years is around \$36.6 million.

D. Miscellaneous Values

In addition to the previously mentioned questions, recorded by the survey, we asked the participants a few miscellaneous questions to help us get a comprehensive picture of the value of the Sanctuary. Miscellaneous questions include things such as the sentimental value of keeping the Sanctuary, the value of the Sanctuary as a status symbol for the school, the appeal of the Sanctuary to incoming students and the resulting increased admissions yield, and the Sanctuary as a habitat for wildlife. We removed any outliers in survey responses using the 2 standard deviation rule, then calculated the average value from survey results for each of the categories, listed in the table below:

How much would the sentimental value that you place on the Sanctuary be worth: how much would you pay per year to keep the Sanctuary as is (ie. would you be sad if it was torn down?) (\$ value per year)	\$689.68/(student*year)
When applying to the school did the Sanctuary do anything to increase the appeal or the value of the school to you? Can you quantify this in a number? (ex would be willing to pay __ amount more for Andover because it has the Sanctuary)	\$183.64/(student*year)
How much would you pay for the Sanctuary to continue to act as a space for the wildlife it contains (currently comprising deer, foxes, geese, swans, and woodland songbirds)?	\$1629.28
How much would you currently pay for the Sanctuary for the ecosystem services it provides (air purification, water collection, climate recreation, etc)?	\$485.48

a. Sentimental Value & Admissions Yield

The sentimental and aesthetic appeal of having a natural space on campus could have significant personal value to many students. To calculate the total sentimental value to all Andover students per year, we assume a student population of 1200 students and apply the following formula:

$$\$689.68 \times 1200 = \$827,616 \text{ per year}$$

To calculate the total increased admissions yield value to Andover per year, we assume that the number of incoming students each year is 300 students and apply the following formula:

$$\$183.64 \times 300 = \$55,092 \text{ per year}$$

To calculate the cumulative sentimental and admissions yield increase value over the next 50 years, we take the sum of the two values discounted per year for 50 years

$$\text{Net Present Value} = \sum_{t=1}^{50} \frac{\$(827,616 + 55,092)}{(1.025)^t} = \$25,035,635.32$$

b. Wildlife Habitat & Ecosystem Services

Our survey asks for a one-time value of how much students would pay for the Sanctuary to continue working as a wildlife habitat and its ecosystem services, therefore the total value of the Sanctuary as a wildlife habitat and for the ecosystem services it provides does not need to be discounted, and instead only needs to be multiplied by the number of Andover students. Assuming the total number of Andover students is 1200, this value is given by the formula below:

$$$(485.48 + 1629.28) \times 1200 = \$2,537,712$$

The total value of miscellaneous elements and services the Sanctuary provides is:

$$\$2,537,712 + \$25,035,635.32 = \$27,573,347.32$$

Conclusion

To conclude, we will conduct a CBA of selling the Sanctuary (immediate value) versus keeping it (long term value). Below are the summary tables of the two values, with long-term values already discounted:

Immediate Value

Land Value	\$36.4 million
Timber Value	\$61,306
Total Immediate Value	\$36.5 million

Long Term Value

Pollution Removal Value	\$1,016,238.64
Carbon Removal Value	\$1,446,177
Recreational Use Value	\$36,558,792.86
Miscellaneous Value	\$27,573,347.32
Total Long Term Value	\$66,594,555.82

In order to conduct a CBA of selling versus keeping the Sanctuary, we calculate the following ratio:

$$\text{Cost Benefit Ratio} = \frac{\text{Total Immediate Value}}{\text{Total Long Term value}} = .55$$

Because the ratio is much smaller than 1, it indicates that the Sanctuary provides more value than selling it. However, our group is skeptical of this assessment for a few reasons. The most obvious one comes from our survey results. Although our survey was sent to around 300 people, we only managed to get around 35 responses and had to remove outliers, which resulted in even less data points. This data was purely subjective and, one could argue, not representative of the entire school. Although we tried to send the survey to a sample representative of our own school (through varied dorms, sports teams, EBI groups, etc), the small sample size doesn't allow us to extrapolate to the larger population. Additionally, we believe the population sampled may not be representative of the entire school because a large portion of the respondents are on the track team, which utilizes the Sanctuary for recreational purposes (running) heavily.

This is important because the majority of the Total Long Term Value depended on the results of the survey (both the Recreation Use and Miscellaneous Values, \$36,558,792.86 and \$27,573,347.32, respectively), so when the survey is not representative of the entire population, these numbers are not necessarily correct, affecting our CBA and conclusion. Additionally, the Andover campus has many other spaces to achieve the same recreational means we surveyed on (e.g., the track for running, art studios for creativity, classrooms for learning, etc.), meaning the sanctuary is not necessary in achieving these recreational means. Taking this sampling error and access other recreational spaces into account and looking at the other parts which make up the Total Long Term Value (Pollution and Carbon Removal, \$1,016,238.64 and \$1,446,177, respectively), we would advise the school to sell the Sanctuary because the Immediate Value is substantial and could be invested into other missions of education or environmental initiatives.

Although we recognize our Land and Timber values were based on estimations and national averages, and are therefore not perfectly accurate, we have more confidence in the accuracy of our Total Immediate Value calculations because they relied on less subjective data.

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